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Kim

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(54) **CONNECTOR WITH WATER INGRESS PROTECTION COVER**

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(30) **Foreign Application Priority Data**

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H01R 24/00 (2011.01)
H01R 13/52 (2006.01)

(52) **U.S. Cl.**
CPC ***H01R 13/5213*** (2013.01); ***H01R 24/00*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

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Primary Examiner — Tho D Ta
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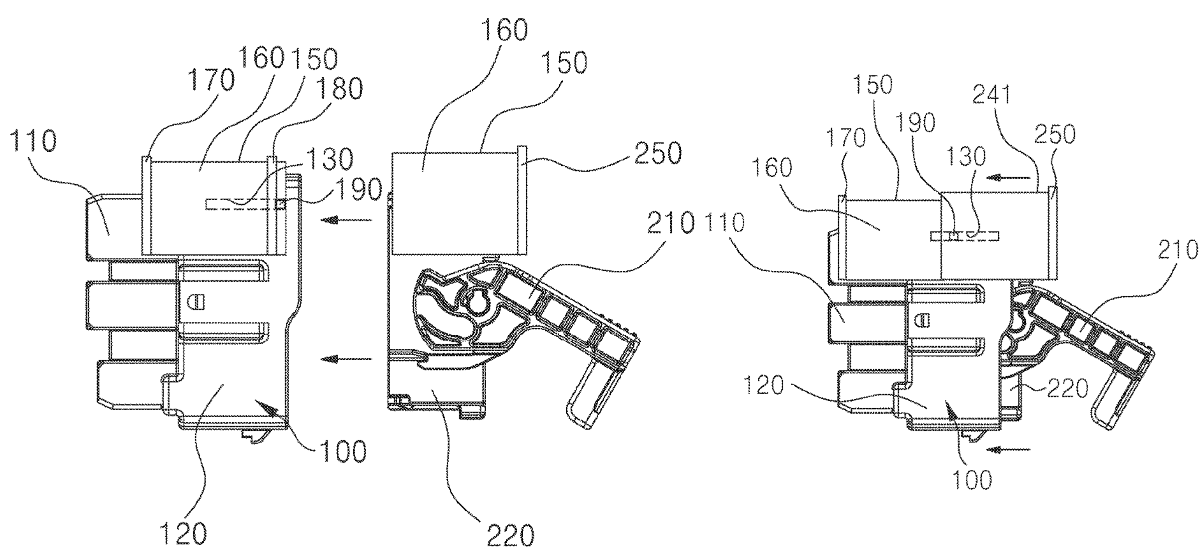
(57) **ABSTRACT**

An embodiment connector with a water ingress protection cover includes a female connector including a housing having a pair of guide holes extended along a first direction at both sides thereof and a first body extended from the housing, a male connector, a first cover mounted to an upper part of the female connector to be movable along the pair of guide holes, and a second cover mounted to an upper part of the male connector. The first cover is configured to move in the first direction to a position over the first body by the second cover or the male connector during coupling of the male connector to the female connector, and the second cover is positioned over the housing and coupled to the first cover in a state in which the male connector is coupled to the female connector.

20 Claims, 17 Drawing Sheets



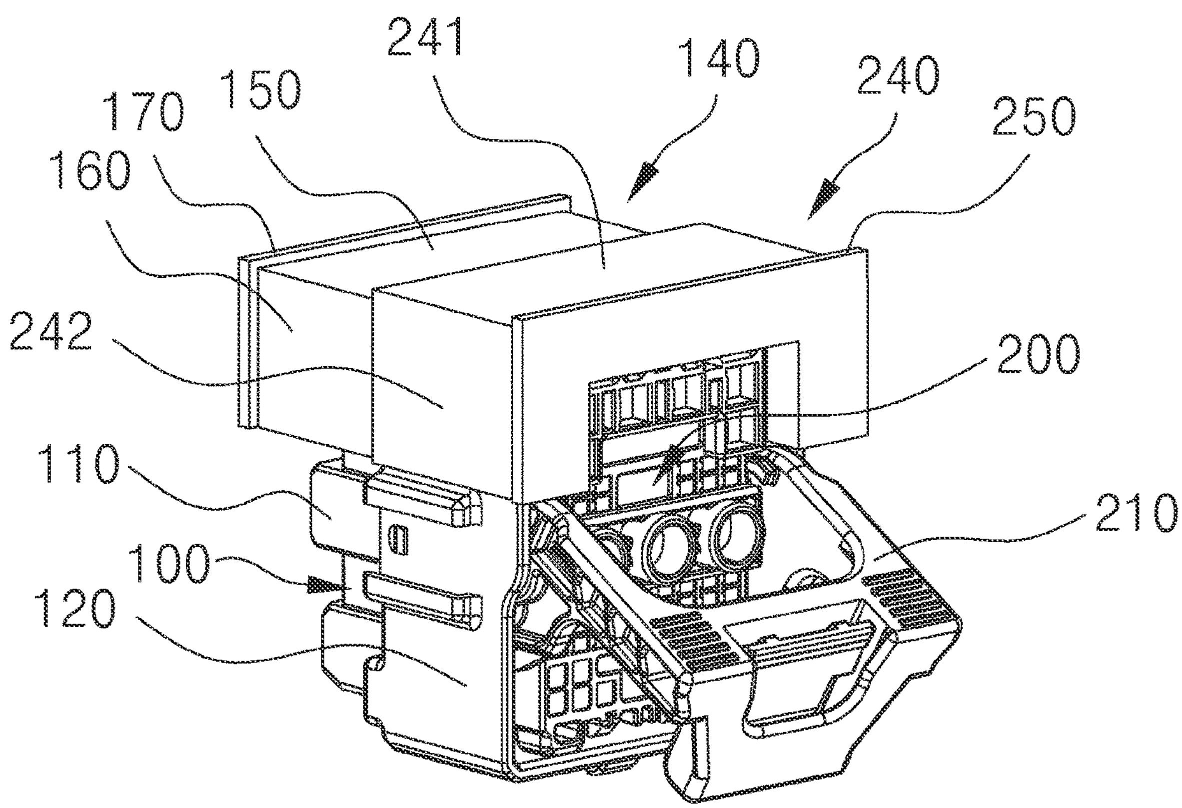


FIG. 1

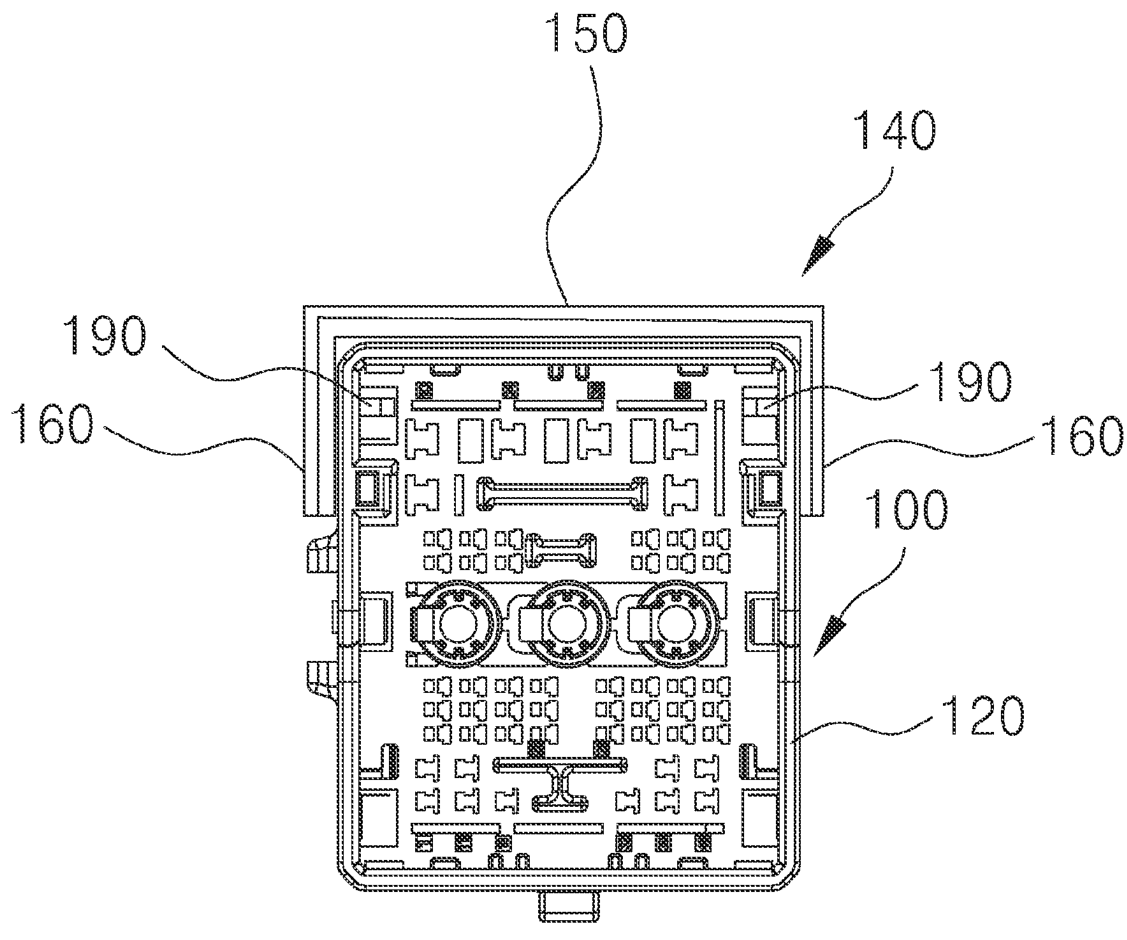


FIG. 2

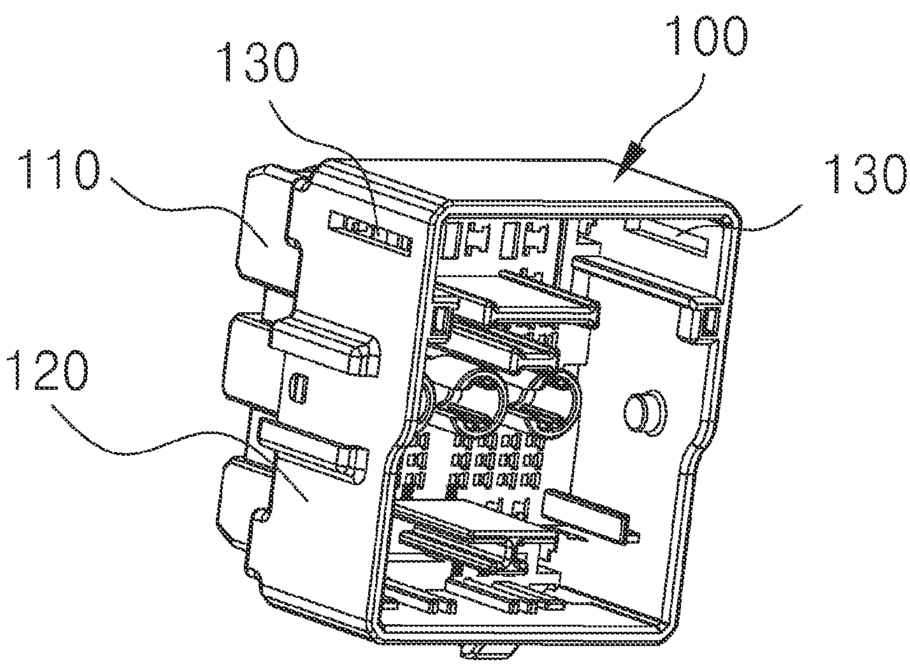


FIG. 3

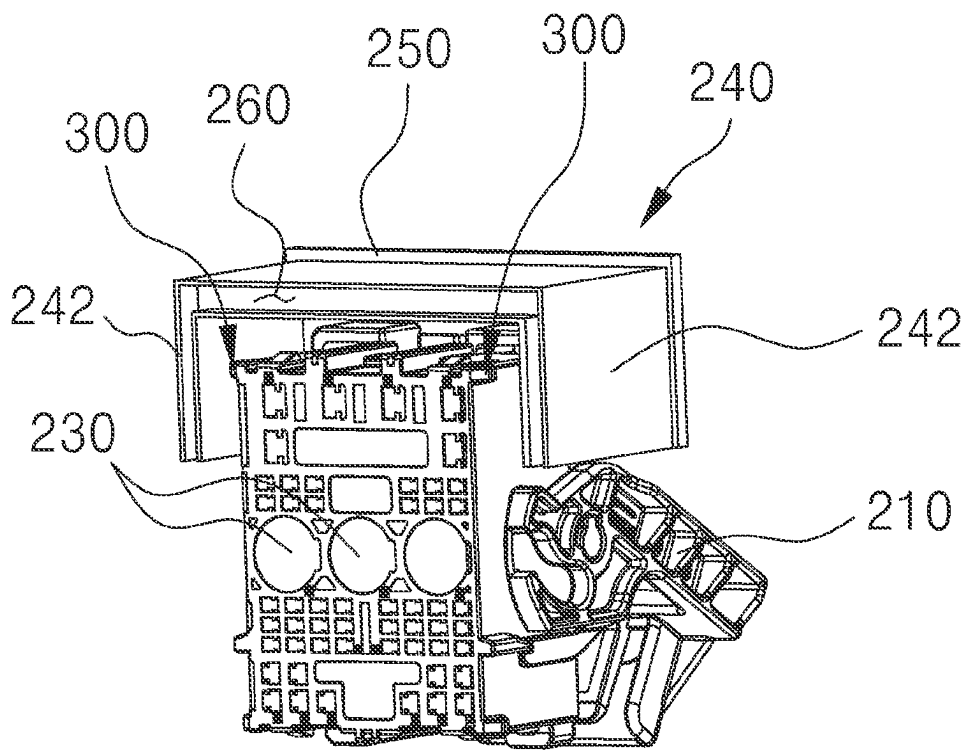


FIG. 4

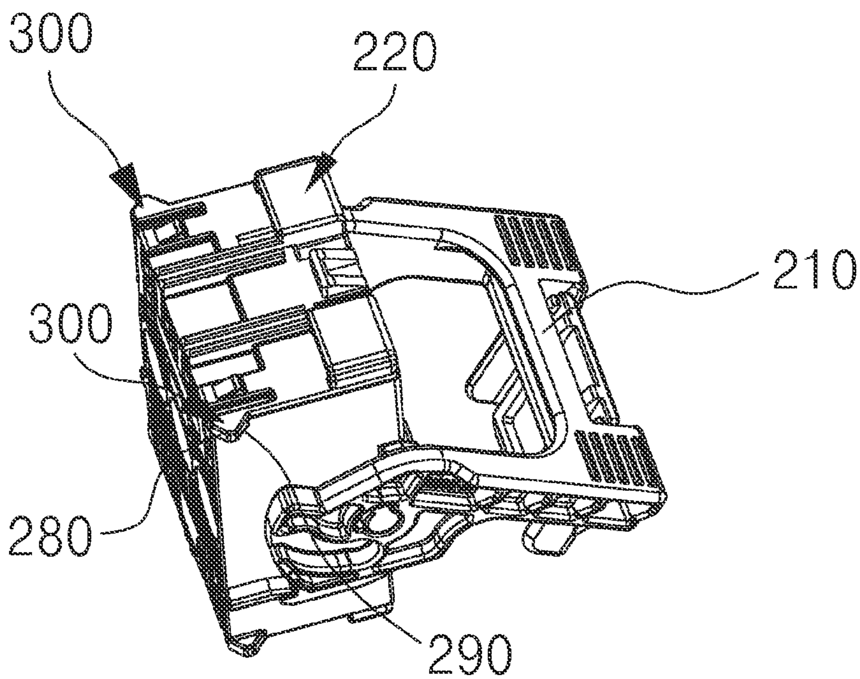


FIG. 5

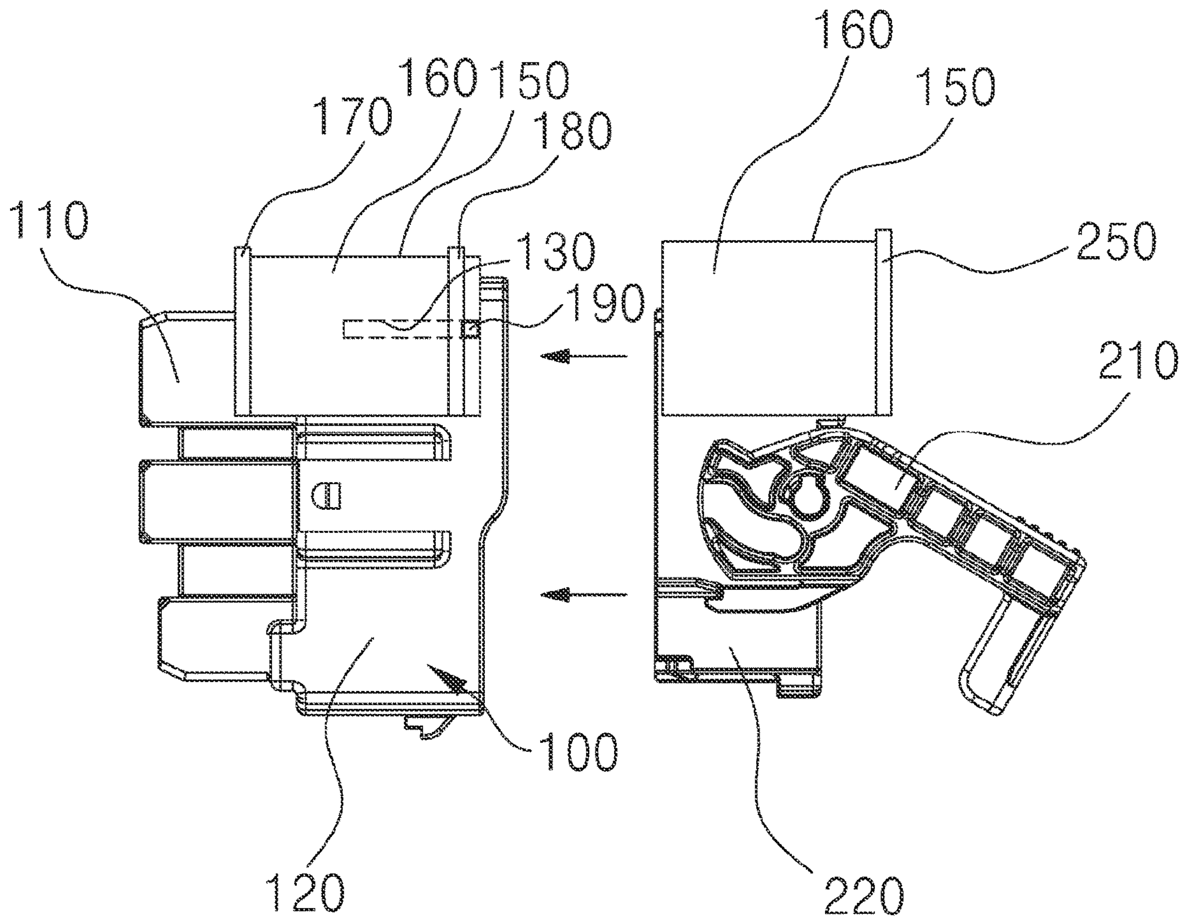


FIG. 6

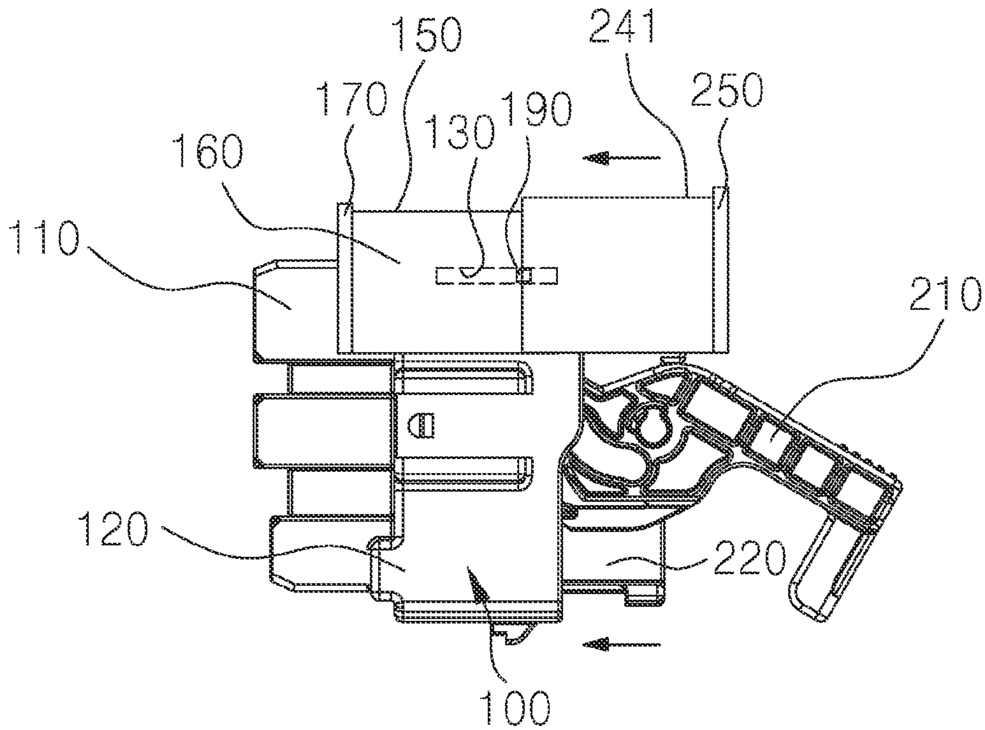


FIG. 7

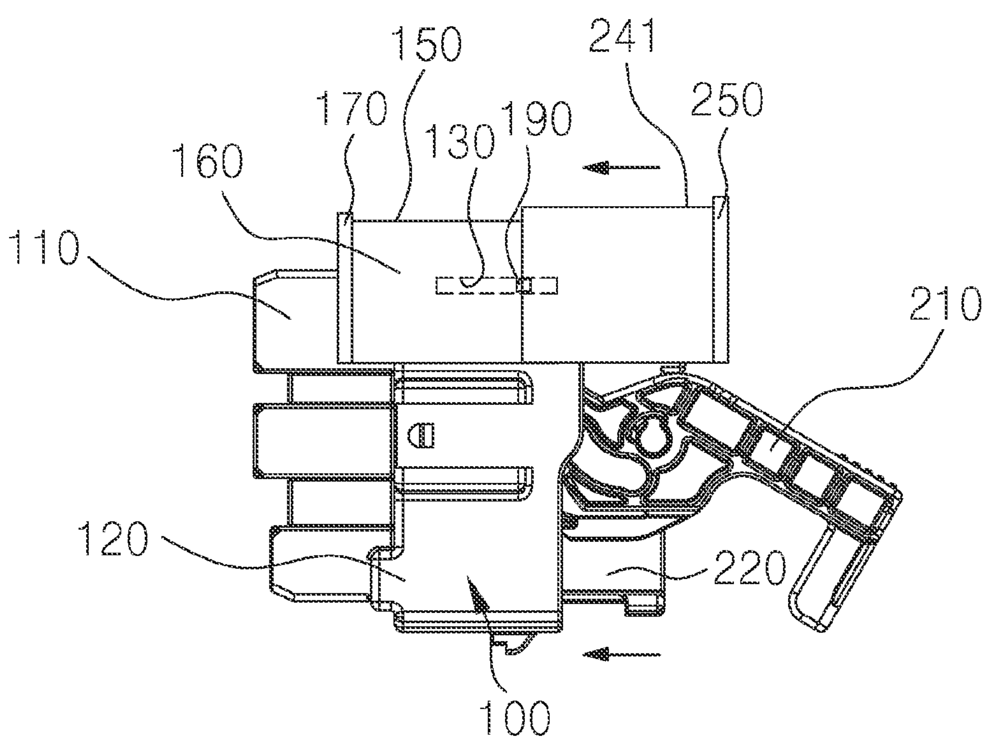


FIG. 8

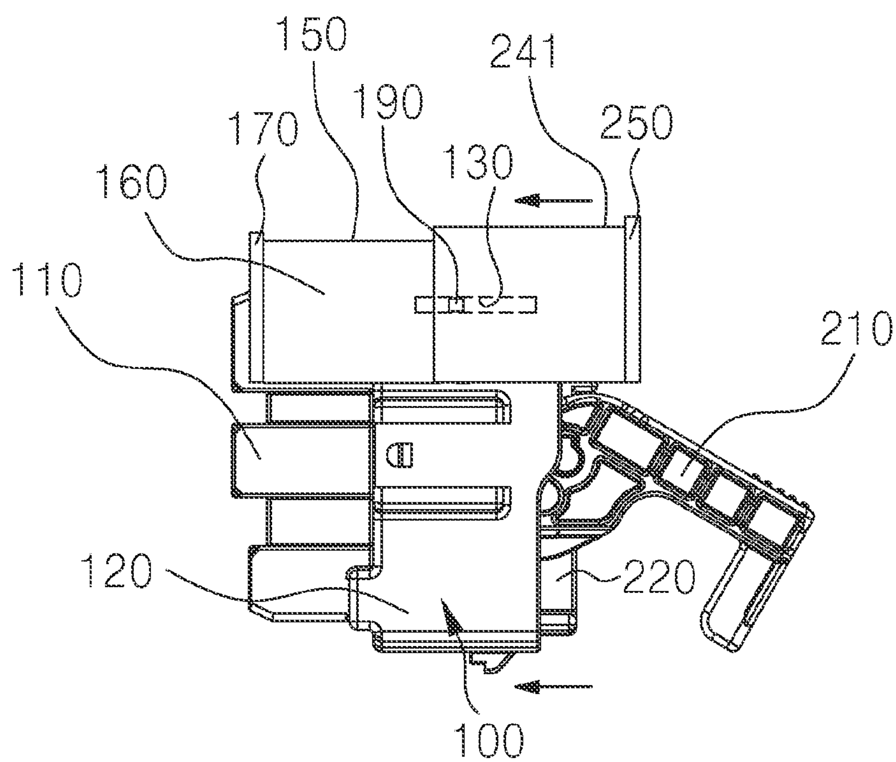


FIG. 9

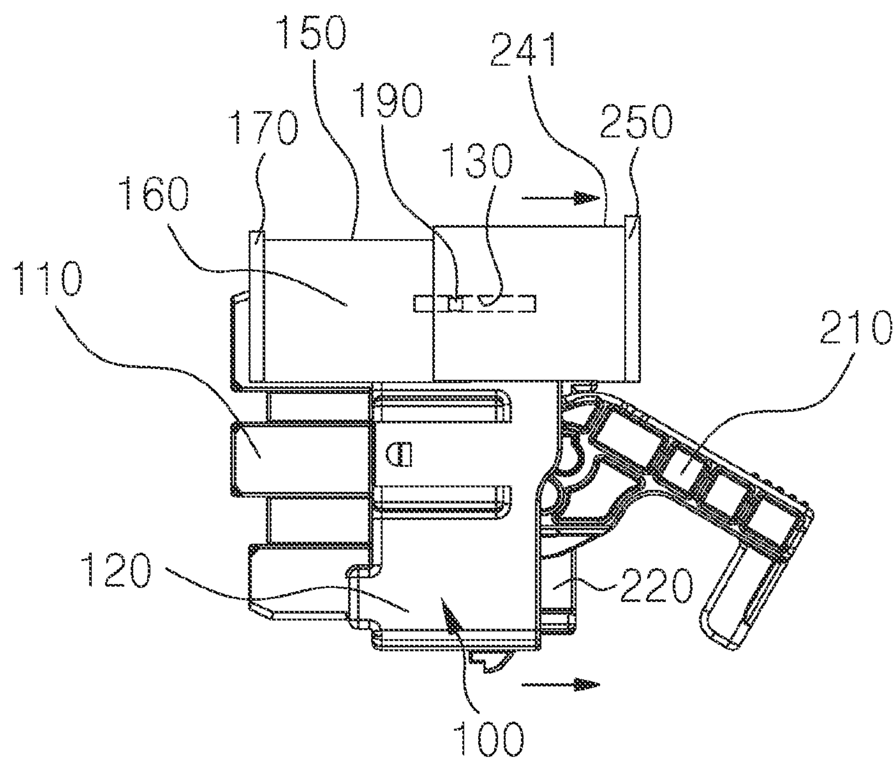


FIG. 10

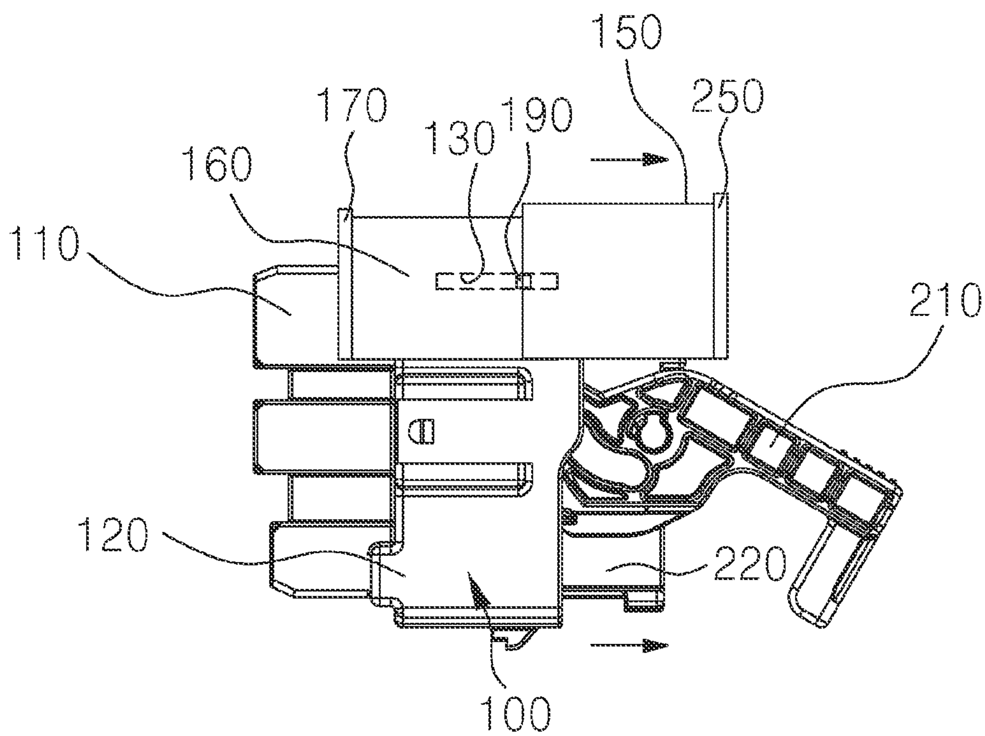


FIG. 11

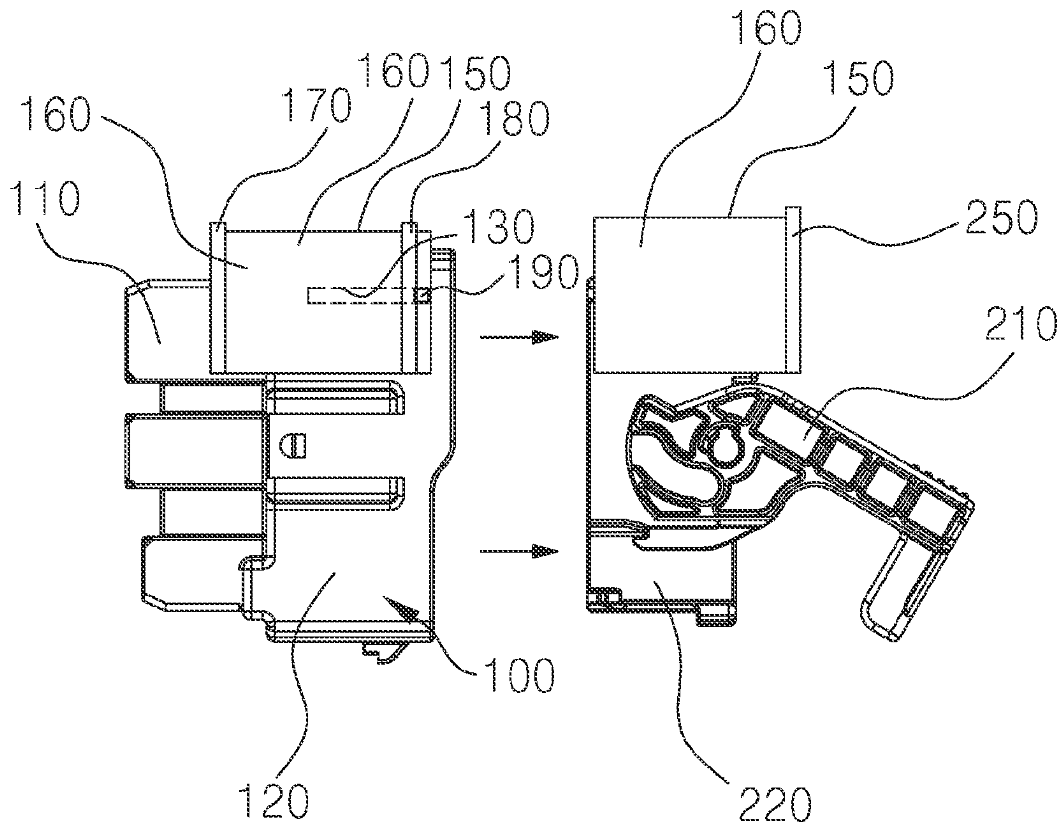


FIG. 12

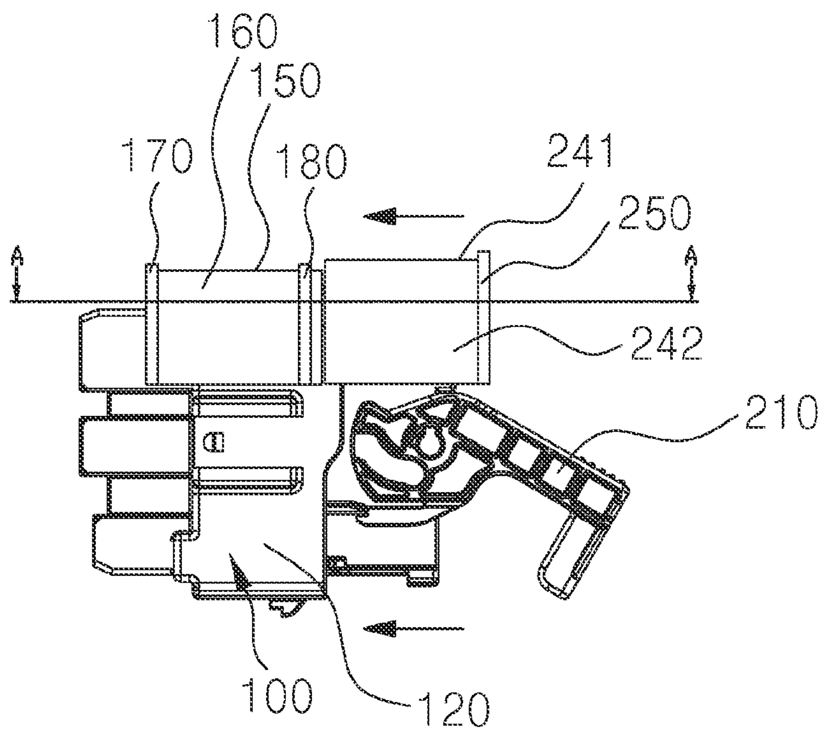


FIG. 13

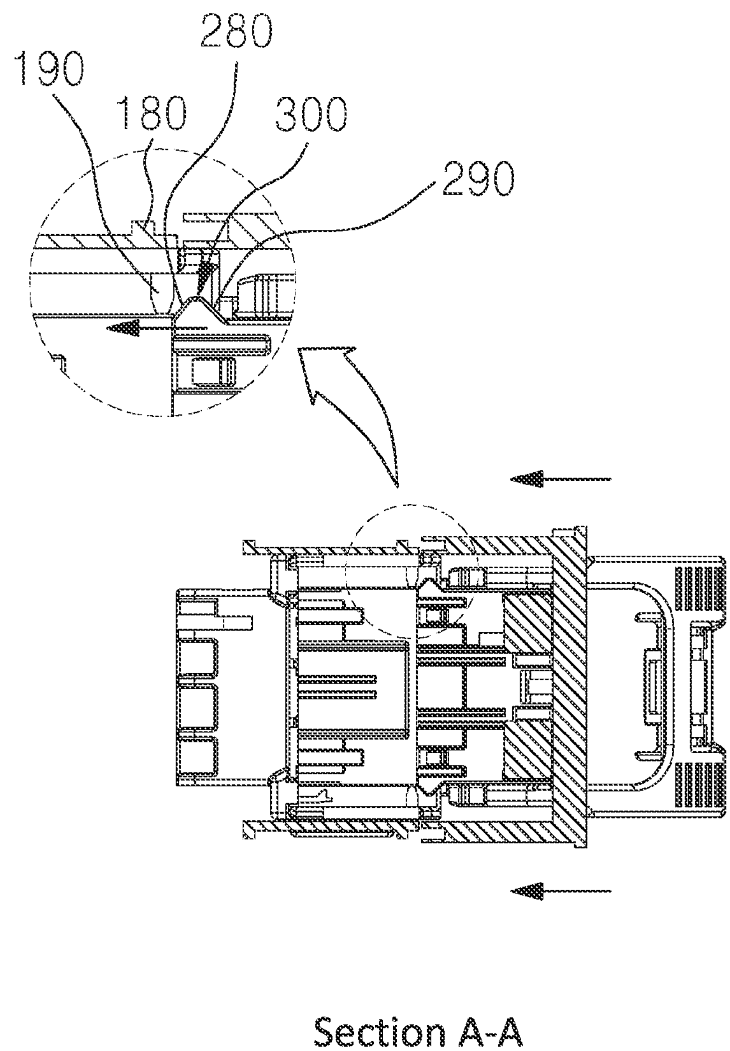


FIG. 14

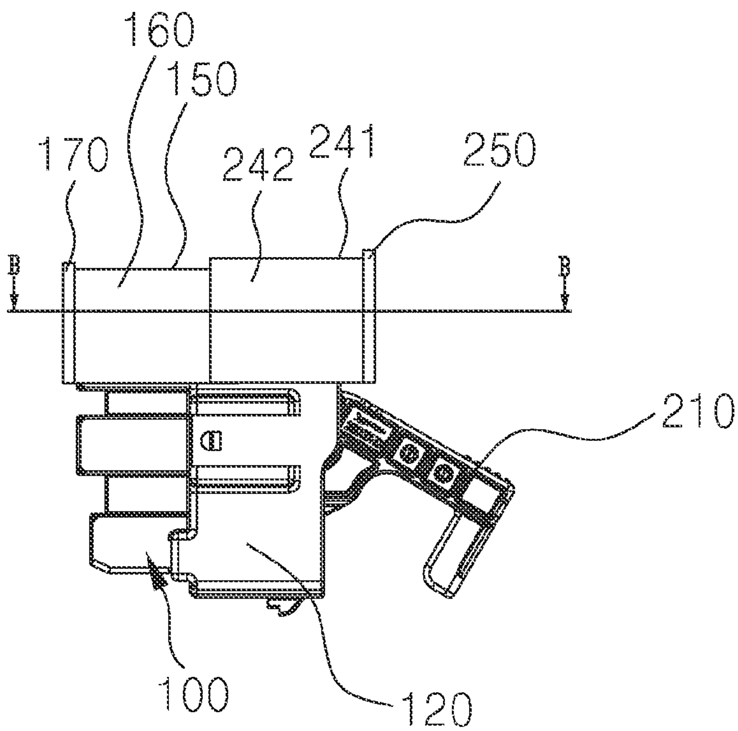
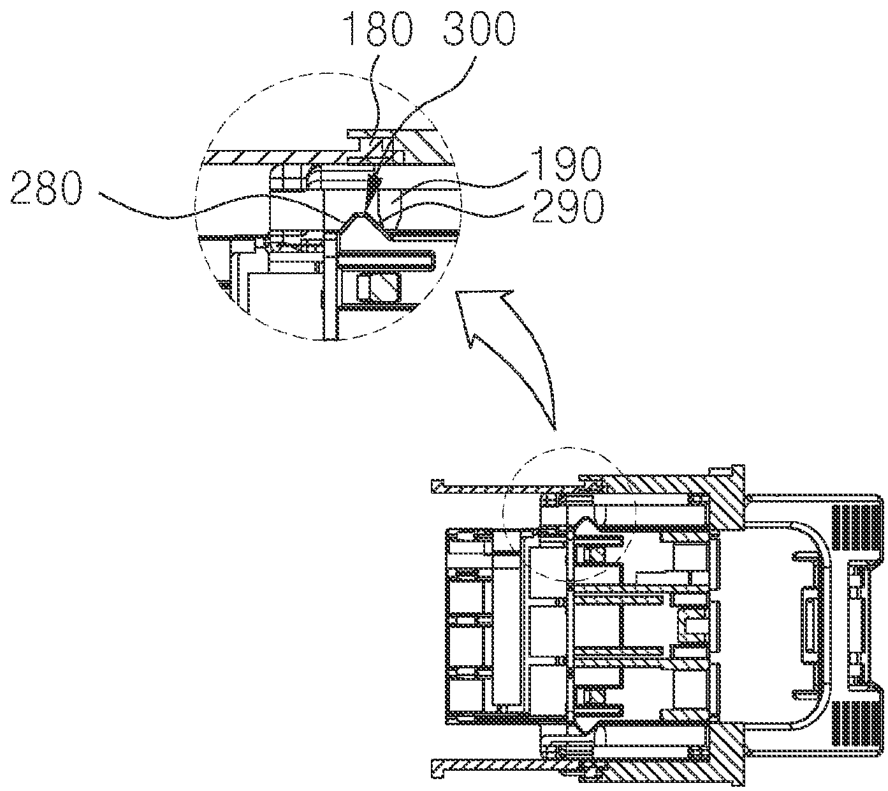


FIG. 15



Section B-B

FIG. 16

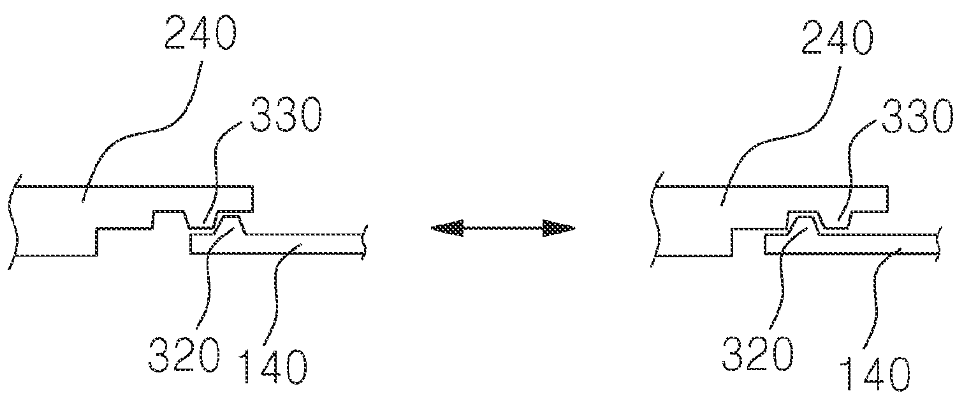


FIG. 17

CONNECTOR WITH WATER INGRESS PROTECTION COVER

CROSS-REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of Korean Patent Application No. 10-2022-0130635, filed on Oct. 12, 2022, which application is hereby incorporated herein by reference.

TECHNICAL FIELD

The present invention relates to a connector with a water ingress protection cover.

BACKGROUND

In recent years, connectors for connecting electronic components to each other have been widely used in vehicles due to the electrification trend.

Connectors are widely used not only outside the vehicle but also inside the vehicle, and there is a very rare chance for water to get into the vehicle, but water or chemicals may fall on top of a connector due to a work such as tinting on the inner surface of the windshield or side glasses or a moisture-resistant coating.

According to the prior art, a multi-box is used to cover a connector to prevent water from entering into the connector.

According to the prior art, multi-boxes of various dimensions are required to cover various connectors, thereby causing a problem of cost increase.

In addition, the above-described operations are regularly performed when water falls on an upper part of the connector, such that only the upper part of the connector can be blocked from water drops, but the multi-box closes the entire connector occupying a large amount of space in the vehicle, thereby causing difficulties in assembling vehicle parts, deteriorating the degree of freedom in designing the vehicle, etc.

Therefore, there is a need for a method to solve problems in the art.

It should be understood that the above-described background technology is only for the purpose of enhancing understanding of the background of embodiments of the present invention and does not mean that the background technology corresponds to a prior technology that is already known to those skilled in the art.

SUMMARY

The present invention relates to a connector with a water ingress protection cover. Particular embodiments relate to a connector having a cover for blocking water falling from an upper part of a connector.

Various embodiments of the present disclosure provide a connector having a first cover and a second cover which are respectively provided on an upper part of a male connector and a female connector and having a new type of a water ingress protection cover which protects the upper parts of the male connector and the female connector from falling water when the male connector and the female connector are coupled.

According to at least one embodiment of the present disclosure, the connector with a water ingress protection cover comprises a female connector including a housing having a pair of guide holes extended along a first direction at both sides thereof and a first body extended from the housing, a male connector, a first cover mounted to an upper part of the female connector movably along the pair of guide holes and a second cover mounted to an upper part of the male connector, wherein the first cover is moved in the first direction to a position over the first body by the second cover or the male connector when the male connector is coupled to the female connector, and the second cover is positioned over the housing with being coupled to the first cover when the male connector is coupled to the female connector.

In at least one embodiment of the present disclosure, the first cover is moved to a position over the housing by the second cover or the male connector when the male connector is separated from the female connector.

In at least one embodiment of the present disclosure, the first cover comprises a first upper surface and a first side surface extending downward from both ends of the first upper surface, and the second cover comprises a second upper surface and a second side surface extending downward from both ends of the second upper surface.

In at least one embodiment of the present disclosure, a first protrusion is formed at one end in the first direction and a second protrusion is formed at the other end on outer surfaces of the first upper surface and the first side surface, and a third protrusion is formed at one end in the first direction on outer surfaces of the second upper surface and the second side surface.

In at least one embodiment of the present disclosure, the second protrusion is formed at a position a predetermined distance apart from the other end in the first direction.

In at least one embodiment of the present disclosure, the second cover has a protrusion groove which the second protrusion is inserted in.

In at least one embodiment of the present disclosure, the connector further comprises a coupling mechanism configured to couple the male connector and the first cover.

In at least one embodiment of the present disclosure, the coupling mechanism comprises a first coupling element formed on the male connector or the second cover and a second coupling element formed in the first cover.

In at least one embodiment of the present disclosure, the first coupling element comprises a bump comprising a first inclined surface formed at one side thereof and a second inclined surface formed at the other side thereof, and the second coupling element comprises a pair of protrusions protruding from inner surfaces of the first cover and coupled to the pair of guide holes.

In at least one embodiment of the present disclosure, the first coupling element comprises a first protrusion formed on an outer surface of the first cover, and the second coupling element comprises a second protrusion formed on an inner surface of the second cover.

When the male connector and the female connector are coupled, the first cover and the second cover are coupled so as to achieve an effect of preventing water from entering into the connector.

Embodiments of the present disclosure are not limited to the above-mentioned embodiments, and other embodiments not mentioned will be clearly understood from the following description by those skilled in the art to which the present disclosure pertains.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a drawing illustrating a connector with a water ingress protection cover according to an embodiment of the present disclosure.

FIG. 2 is a drawing illustrating a female connector of the connector with a water ingress protection cover according to an embodiment of the present disclosure.

FIG. 3 is a drawing illustrating a female connector in which a first cover is separated from a connector with a water ingress protection cover according to an embodiment of the present disclosure.

FIG. 4 is a drawing illustrating a male connector of a connector with a water ingress protection cover according to an embodiment of the present disclosure.

FIG. 5 is a drawing illustrating a male connector in which a second cover is separated from a connector with a water ingress protection cover according to an embodiment of the present disclosure.

FIG. 6 is a drawing illustrating a state in which a male connector is coupled to a female connector in a connector with a water ingress protection cover according to an embodiment of the present disclosure.

FIG. 7 is a drawing illustrating a state in which a first cover and a second cover are coupled to each other when a male connector and a female connector are coupled to each other in a connector with a water ingress protection cover according to an embodiment of the present disclosure.

FIG. 8 is a drawing illustrating a state in which a first cover is moved by a second cover in a connector with a water ingress protection cover according to an embodiment of the present disclosure.

FIG. 9 is a drawing illustrating a state in which a first cover is completely moved from a connector with a water ingress protection cover according to an embodiment of the present invention.

FIG. 10 and FIG. 11 are drawings illustrating a state in which a male connector and a female connector are separated from a connector with a water ingress protection cover according to an embodiment of the present invention and a first cover is moved by the male connector.

FIG. 12 is a drawing illustrating a state in which a male connector is separated from a female connector in a connector with a water ingress protection cover according to an embodiment of the present invention.

FIG. 13 is a drawing illustrating a state in which a male connector is coupled to a female connector in a connector with a water ingress protection cover according to an embodiment of the present invention.

FIG. 14 is a cross-sectional view taken along line A-A of FIG. 13.

FIG. 15 is a drawing illustrating a state in which a first cover is completely moved from a connector with a water ingress protection cover according to an embodiment of the present invention.

FIG. 16 is a cross-sectional view taken along line B-B of FIG. 15.

FIG. 17 is a drawing illustrating a state in which a first cover and a second cover are coupled and uncoupled.

The following reference identifiers may be used in connection with the figures to describe exemplary embodiments of the present invention.

100: Female connector	110: First body
120: Housing	130: Guide hole
140: First cover	150: First upper surface
160: First side surface	170: First protrusion
180: Second protrusion	190: Protrusion
200: Male connector	210: Handle
220: Second body	230: Terminal hole
240: Second cover	241: Second upper surface

-continued

242: Second side surface	250: Third protrusion
260: Protrusion groove	270: First coupling element
280: First inclined surface	290: Second inclined surface
300: Bump	310: Second coupling element
320: First protrusion	330: Second protrusion

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

Since embodiments of the present disclosure may be modified in various ways and may have various embodiments, specific embodiments will be illustrated and described in the accompanying drawings. However, this is not intended to limit embodiments of the present invention to specific embodiments, and it should be understood that embodiments of the present invention include all modifications, equivalents, and replacements included on the idea and technical field of embodiments of the present invention.

Terms including ordinals such as "first," "second," etc., may be used to describe various elements, but the components are not limited by these terms. The terms are used only for the purpose of distinguishing one component from another component.

The term "and/or" may be used to include any combination of a plurality of items to be included. For example, "A and/or B" includes all three cases such as "A," "B," and "A and B."

When a component is mentioned as being "connected" or "linked" to another component, the component may be directly connected or linked to the other component, but it should be understood that another component may exist in between. On the other hand, when it is mentioned that a component is "directly connected" or "directly linked" to another component, it should be understood that another component does not exist in between.

In the description of embodiments, it will be understood that a layer (film), region, pattern, or structure is formed "on" or "under" another layer (film), region, pad, or pattern, either directly or through intervening layers. The reference to "up/on" or "down/under" is based on the elements shown in the drawings for convenience of explanation, is used to show the relative positional relationship between the components for convenience of explanation, and should not be understood as limiting the actual positions of the components. For example, "above B" is merely an indication that B is shown above A in the drawings, unless otherwise stated or A should not be positioned above B due to the property of A or B, B may be positioned below A or B, and A may be disposed adjacently in the actual product or the like.

In addition, a density or a size of each layer (film), region, pattern, or structure in the drawings may be modified for clarity and convenience of the description, and thus does not entirely reflect the actual size.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of embodiments of the present disclosure. Singular expressions include plural expressions, unless the context clearly indicates otherwise. In the present application, it should be understood that the term "include" or "have" indicates that a feature, a number, a step, an operation, a component, a part, or a combination thereof described in the specification is present, but does not exclude the possibility of existence or addition of one or more other features, numbers, steps, operations, components, parts, or combinations thereof in advance.

Unless otherwise defined, all terms used herein, including technical or scientific terms, have the same meaning that is generally understood by those skilled in the art to which embodiments of the present invention pertain. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or a strictly formal meaning unless expressly so defined herein.

Hereinafter, exemplary embodiments will be described in detail with reference to the accompanying drawings, in which like reference numerals refer to like or corresponding elements throughout the figures, and repetitive descriptions thereof will be omitted.

The connector with a water ingress protection cover according to embodiments of the present disclosure comprises a female connector **100**, a male connector **200**, a first cover **140**, and a second cover **240**.

The female connector **100** comprises a first body **110** and a housing **120** formed at one side of the first body **110**, a pair of guide holes **130** are formed at both sides of the housing **120**, and a protrusion **190**, which will be described later, is inserted into the guide hole **130** to be guided by sliding. A plurality of first terminals (not shown) may be fixed to the first body **110** by an insert molding process with first ends thereof exposed to the housing **120**.

The male connector **200** includes a second body **220** having a size to be inserted into the housing **120**.

The second body **220** includes a plurality of terminal holes **230** where a plurality of first terminals exposed to the housing **120** are inserted, and a second terminal (not shown) contacting the first terminal is provided in the second body **220** by insert molding process.

The male connector **200** is provided with a handle **210** to easily separate the second body **220** inserted into the housing **120**.

The first cover **140** is mounted at the upper part of the female connector **100** to be slidable in a first direction, i.e., the coupling direction of the male connector **200**.

The first cover **140** includes a first upper surface **150** and a pair of first side surfaces **160** extending downward from both ends of the first upper surface **150**.

In the first upper surface **150** and the first side surfaces **160**, as shown in FIGS. **6** and **12**, a first protrusion **170** is formed at one end and a second protrusion **180** is formed at the other end.

The second protrusion **180** is formed at a position apart from the other end at a preset distance.

Referring to FIG. **4**, one side of the second cover **240** is mounted on the upper part of the male connector **200**, specifically fixed on the second body **220**.

The second cover **240** comprises a second upper surface **241** and a pair of second side surfaces **242** extending downward from both ends of the second upper surface **241**.

A third protrusion **250** is formed on one end of the second upper surface **241** and the second side surfaces **242**.

The second cover **240** has a protrusion groove **260** formed on the inner surface thereof so that the second protrusion **180** is inserted therein.

The first protrusion **170** through the third protrusion **250** guide water falling on the first upper surface **150** and the second upper surface **241** to flow to the first side surface **160** and the second side surface **242**.

The connector comprises a coupling mechanism for coupling the male connector **200** and the first cover **140**.

The coupling mechanism includes a first coupling element **270** and a second coupling element **310**.

The first coupling element **270** is formed outside the male connector **200**.

As shown in FIG. **14**, the first coupling element **270** comprises a bump **300** comprising a first inclined surface **280** formed on one side thereof and a second inclined surface **290** formed on the other side thereof to be symmetrical to the first inclined surface **280**.

The second coupling element **310** is formed on the inner surface of the first side surface **160**. The second coupling element **310** includes the protrusion **190** protruding from the inner surface of the first side surface **160**.

The coupling mechanism may be applied in another embodiment, and the first coupling element **270** may include the first protrusion **320** formed on the outer surface of the first cover **140**, and the second coupling element **310** may include the second protrusion **330** formed on the inner surface of the second cover **240**.

The first protrusion **320** and the second protrusion **330** may be coupled as shown in FIG. **17**.

Next, a process of coupling the male connector **200** to the second cover **240** after the first cover **140** is moved when the male connector **200** is coupled to the female connector **100** will be described in reference to FIGS. **6** to **16**.

When the male connector **200** is coupled to the female connector **100** as illustrated in FIG. **6**, the first cover **140** and the second cover **240** are coupled to each other while in contact as illustrated in FIG. **7**.

As shown in FIG. **8**, when the second body **220** of the male connector **200** is inserted into the housing **120** of the female connector **100**, the second cover **240** moves the first cover **140** toward the first body **110**.

When the second body **220** is inserted into the housing **120**, as shown in FIG. **9**, the first cover **140** is located above the first body **110**, and the second cover **240** is located above the housing **120**. Accordingly, the first cover **140** and the second cover **240** cover the entire upper parts of the male connector **200** and the female connector **100**.

When the second body **220** is inserted into the housing **120**, if the first cover **140** is guided by the guide hole **130** and moved with a predetermined distance, the first inclined surface **280** and the protrusion are in contact with each other as shown in FIG. **14**.

In this case, when the second body **220** of the male connector **200** is further inserted into the housing **120**, the protrusion moves along the first inclined surface **280** and moves toward the second inclined surface **290** as shown in FIG. **16**.

When the male connector **200** is separated from the female connector **100**, the male connector moves the first cover **140** in the upper direction of the housing **120** as shown in FIGS. **10** to **12**.

In detail, when the second body **220** of the male connector **200** is separated from the housing **120**, the second inclined surface **290** contacts the protrusion **190**, and when the bump **300** having the second inclined surface **290** formed thereon is moved, the protrusion **190** is moved along the second inclined surface **290** to move the first cover **140** to the upper position of the lower housing **120**.

When the movement of the first cover **140** to the upper part of the housing **120** is completed, the male connector **200** and the female connector **100** may be completely separated as the protrusion **190** moves from the second inclined surface **290** to the first inclined surface **280**.

In another embodiment, although not shown in the drawings, when the male connector **200** is coupled to the female connector **100**, the moved first cover **140** may be slid to the upper part of the housing **120** by the elastic mechanism without the above-mentioned coupling mechanism when the male connector **200** is separated from the female connector **100**. The elastic mechanism may be a coil spring.

Although the above description has been focused on exemplary embodiments, these are merely examples and embodiments of the present invention are not limited thereto, and it will be understood by those skilled in the art that various modifications and applications that are not illustrated above may be made without departing from the essential characteristics of the present embodiments. For example, each component specifically shown in the embodiments may be modified and implemented. Further, the differences related to the modifications and applications should be interpreted as being included in the scope of embodiments of the present invention defined in the appended claims.

What is claimed is:

1. A connector with a water ingress protection cover, the connector comprising:

- a female connector comprising a housing having a pair of guide holes extended along a first direction at both sides thereof and a first body extended from the housing;
- a male connector;
- a first cover mounted to an upper part of the female connector to be movable along the pair of guide holes; and
- a second cover mounted to an upper part of the male connector;
- wherein the first cover is configured to move in the first direction to a position over the first body by the second cover or the male connector during coupling of the male connector to the female connector; and
- wherein the second cover is positioned over the housing and coupled to the first cover in a state in which the male connector is coupled to the female connector.

2. The connector of claim **1**, wherein the first cover is configured to be moved to a position over the housing by the second cover or the male connector during separating of the male connector from the female connector.

3. The connector of claim **1**, wherein:

- the first cover comprises:
  - a first upper surface; and
  - first side surfaces extending downward from both ends of the first upper surface; and
- the second cover comprises:
  - a second upper surface; and
  - second side surfaces extending downward from both ends of the second upper surface.

4. The connector of claim **3**, further comprising:

- a first protrusion at a first end in the first direction on outer surfaces of the first upper surface and the first side surfaces;
- a second protrusion at a second end on outer surfaces of the first upper surface and the first side surfaces; and
- a third protrusion at a first end in the first direction on outer surfaces of the second upper surface and the second side surfaces.

5. The connector of claim **4**, wherein the second protrusion is spaced a predetermined distance apart from the second end in the first direction.

6. The connector of claim **4**, wherein the second cover comprises a protrusion groove configured to receive the second protrusion.

7. A connector with a water ingress protection cover, the connector comprising:

- a female connector comprising a housing having a pair of guide holes extended along a first direction at both sides thereof and a first body extended from the housing;
- a male connector;
- a first cover mounted to an upper part of the female connector to be movable along the pair of guide holes; and
- a second cover mounted to an upper part of the male connector;
- a coupling mechanism coupling the male connector and the first cover;
- wherein the first cover is configured to move in the first direction to a position over the first body by the second cover or the male connector during coupling of the male connector to the female connector; and
- wherein the second cover is positioned over the housing and coupled to the first cover in a state in which the male connector is coupled to the female connector.

8. The connector of claim **7**, wherein the coupling mechanism comprises a first coupling element on the male connector or the second cover and a second coupling element in the first cover.

9. The connector of claim **8**, wherein:

- the first coupling element comprises a bump comprising a first inclined surface at a first side thereof and a second inclined surface at a second side thereof; and
- the second coupling element comprises a pair of protrusions protruding from inner surfaces of the first cover and coupled to the pair of guide holes.

10. The connector of claim **8**, wherein:

- the first coupling element comprises a first protrusion on an outer surface of the first cover; and
- the second coupling element comprises a second protrusion on an inner surface of the second cover.

11. The connector of claim **7**, wherein the first cover is configured to be moved to a position over the housing by the second cover or the male connector during separating of the male connector from the female connector.

12. The connector of claim **7**, wherein:

- the first cover comprises:
  - a first upper surface; and
  - first side surfaces extending downward from both ends of the first upper surface; and
- the second cover comprises:
  - a second upper surface; and
  - second side surfaces extending downward from both ends of the second upper surface.

13. The connector of claim **12**, further comprising:

- a first protrusion at a first end in the first direction on outer surfaces of the first upper surface and the first side surfaces;
- a second protrusion at a second end on outer surfaces of the first upper surface and the first side surfaces; and
- a third protrusion at a first end in the first direction on outer surfaces of the second upper surface and the second side surfaces.

14. The connector of claim **13**, wherein the second protrusion is spaced a predetermined distance apart from the second end in the first direction.

15. The connector of claim **13**, wherein the second cover comprises a protrusion groove configured to receive the second protrusion.

16. A connector with a water ingress protection cover, the connector comprising:

a female connector comprising a housing having a pair of guide holes extended along a first direction at both sides thereof and a first body extended from the housing;

a male connector;

a first cover mounted to an upper part of the female connector to be movable along the pair of guide holes, the first cover comprising:

a first upper surface; and

first side surfaces extending downward from both ends of the first upper surface; and

a second cover mounted to an upper part of the male connector, the second cover comprising:

a second upper surface; and

second side surfaces extending downward from both ends of the second upper surface;

wherein the first cover is configured to move in the first direction to a position over the first body by the second cover or the male connector during coupling of the male connector to the female connector;

wherein the second cover is positioned over the housing and coupled to the first cover in a state in which the male connector is coupled to the female connector; and

wherein the first cover is configured to be moved to a position over the housing by the second cover or the male connector during separating of the male connector from the female connector.

17. The connector of claim **16**, further comprising:

a first protrusion at a first end in the first direction on outer surfaces of the first upper surface and the first side surfaces;

a second protrusion at a second end on outer surfaces of the first upper surface and the first side surfaces; and

a third protrusion at a first end in the first direction on outer surfaces of the second upper surface and the second side surfaces.

18. The connector of claim **17**, wherein the second protrusion is spaced a predetermined distance apart from the second end in the first direction.

19. The connector of claim **17**, wherein the second cover comprises a protrusion groove configured to receive the second protrusion.

20. The connector of claim **17**, further comprising a coupling mechanism coupling the male connector and the first cover.

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